DoTS: FAIRly publishing your textual data with the DTS API

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Introduction

In the field of digital humanities, publishing structured textual data in XML/TEI presents significant challenges. The ever-growing number of corpora raises complex questions: how can we ensure the long-term preservation of these corpora? How can we make these texts citable, interoperable, and easily reusable in various contexts?

These issues lie at the heart of the FAIR principles (*Findable*, *Accessible*, *Interoperable*, *Reusable*), which aim to promote the sharing and reuse of research data. The DTS (*Distributed Text Services*) API was specifically designed to address these challenges. Much like the IIIF API has revolutionized image sharing, ”the Distributed Text Services (DTS) Specification defines an API for working with collections of text as machine-actionable data”.

However, the availability of concrete implementations is essential to encourage the adoption and appropriation of DTS by the community. This is precisely where **DoTS** (https://github.com/chartes/dots), an implementation of the DTS API developed in Xquery and supported by the powerful BaseX database engine, comes in.

This poster aims to present DoTS, an open source project founded by the digital infrastructure Biblissima+ (https://projet.biblissima.fr/fr). DoTS is a comprehensive and functional suite of tools for publishing corpora in compliance with the DTS specification, integrating backend, API responses, and frontend for the creation of adaptable websites.

Overview of DoTS and its benefits

Key features of DoTS

DoTS, whose documentation is available online (https://chartes.github.io/dots_documentation/), is a suite of tools developed in Xquery, designed to :

1. **Create and manage projects in BaseX :** DoTS simplifies the setup and management of TEI-based publication projects in BaseX.
2. **Implement a DTS resolver** : Based on the created project, DoTS fully leverages the DTS API (https://distributed-text-services.github.io/specifications/) through its three main endpoints :
   - **Collection :** Lists collection members and provides access to collection or document metadata.
   - **Navigation :** Explores document fragments, accesses their metadata.
   - **Document :** Displays full documents (in XML/TEI or HTML), extracts specific fragments.
3. **Build consultation websites :** With DoTS-vue, a front-end based on the Vue.js framework, users can easily deploy websites where content is directly published using DTS API queries.

Advantages of DoTS

- **Compliance with standard formats :** In line with DTS specifications, HTTP responses from DoTS use JSON-LD for metadata and XML/TEI by default for textual data, ensuring maximum interoperability.
- **Ease of use :** DoTS is designed to be accessible to editors working with TEI corpora. The various cookbooks available in the documentation (https://chartes.github.io/dots_docu-

mentation/cookbook/) demonstrate the main advantages of DoTS :

- **Generic :** It allows users to freely organize documents into collections.
- **Flexible :** Editors can customize metadata to describe collections, documents, or specific passages.
- **Granular :** Passages can be defined according to user needs, enabling precise citation of documents or their fragments.

A Proven tool already in production

DoTS is already in production at the École des chartes – PSL (https://theses.chartes.psl.eu/) and is increasingly adopted, particularly within a growing French-speaking user community. This demonstrates its relevance and effectiveness in supporting Open Science principles and facilitating the publication of textual data.

Conclusion

The DH 2025 conference offers a unique opportunity to showcase this generic tool and establish partnerships to promote the adoption of the DTS API and its DoTS implementation.

A broader adoption of DoTS will :

- Enhance the publication of textual data, ensuring their accessibility, citability, and interoperability.
- Facilitate text alignment and republication, for instance, in multilingual contexts.

By highlighting the strengths of DoTS and the benefits of the DTS API, we aim to contribute to the development of tools that respect the FAIR principles in digital humanities.